higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 4

(12041034)

12 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 7 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

1.1 Substations are no longer using oil circuit breakers, but they are using vacuum and air blast breakers.

State ONE disadvantage of using oil circuit breakers. (1)

1.2 FIGURE 1 below shows a diagram of an indirect dry-cooling system. Study the diagram and label the components marked A to E.

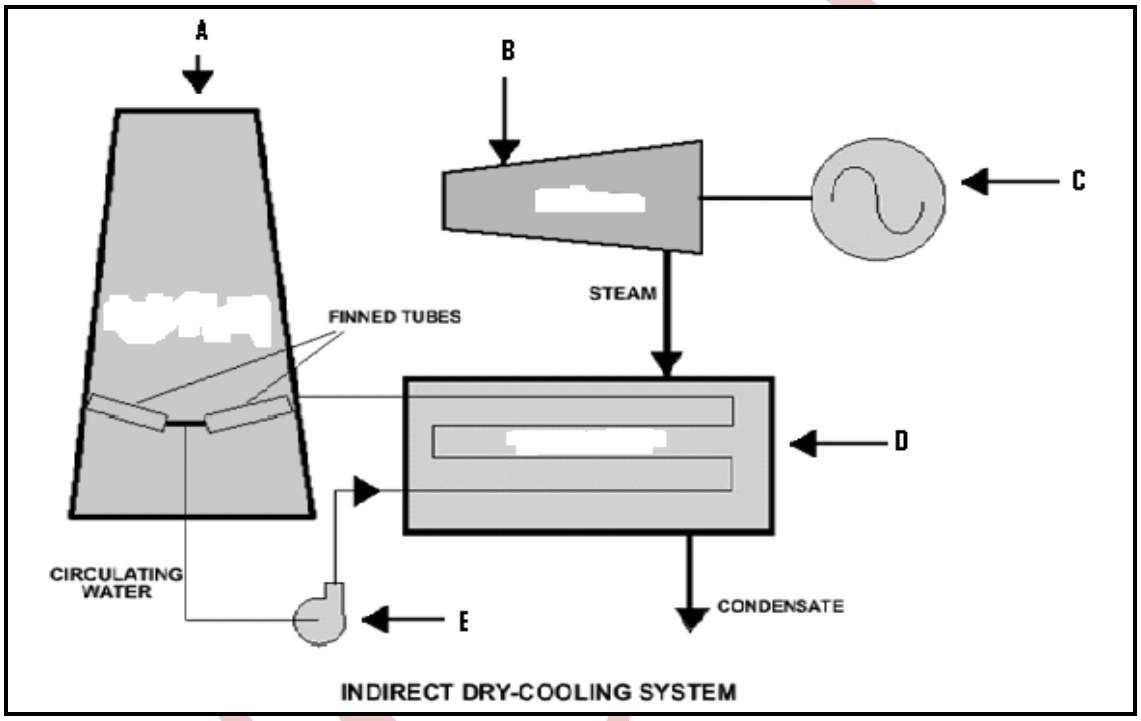


FIGURE 1

(5 × 1) (5)

1.3 Write down the function of each of the following components of a coal-fired power station:

1.3.1 Generator

1.3.2 Furnace

1.3.3 Flue gas stack

1.3.4 Radiator type heat exchanger

1.3.5 Condenser

(5 × 2) (10)

1.4 Name TWO types of insulators used on overhead transmission lines. (2)

1.5 Briefly describe the type of distribution network configuration used in rural areas. (2)

[20]

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QUESTION 2

2.1 Design a three-phase 380 V automatic star/delta motor starter control circuit.
(4 marks for sketch, 2 marks for labels) (6)

2.2 The most common loads used in electrical circuits are inductive and resistive loads.

Briefly describe an inductive load. (2)

2.3 A one-line schematic diagram is used when designing a three-phase circuit.

Name TWO items that must appear on a one-line schematic diagram. (2)

2.4 When the construction work must begin, workers need to comply with worksite procedures.

Give THREE worksite procedures that must be complied with during the construction of circuits. (3)

2.5 Once the design circuit has been completed, it must be evaluated.

Name TWO tests that can be conducted by using a multifunction safety compliance analyser. (2)
[15]

QUESTION 3

3.1 Name ONE electrical parameter required for appropriate design according to National Rationalised Specifications (NRS). (1)

3.2 Name THREE categories of fundamental requirements for a cable when designing a medium-voltage supply network. (3)

3.3 Aluminium conductors reinforced with steel (ACSR) are used on medium overhead transmission lines.

List THREE advantages of using ACSRs. (3)

3.4 Weather influences the design of an electricity distribution network.

Explain how the following weather conditions affect distribution networks:

3.4.1 Wind

3.4.2 Lightning

3.4.3 Salt fog

(3 × 2) (6)

3.5 Name TWO reasons why it is important to know what type of soil is found on a terrain where overhead transmission lines will be constructed. (2)
[15]

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QUESTION 4

4.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.1.1–4.1.10) in the ANSWER BOOK.

4.1.1 More than one electrical diagram is used for domestic installations.

4.1.2 Site plans show the location of electrical systems on every floor.

4.1.3 Hazards caused by electrical equipment may possibly arise as a result of the maintenance of electrical installations.

4.1.4 During risk assessment, you need to identify only hidden hazards.

4.1.5 General Safety Regulations, 1983 should also be consulted when dealing with low-voltage electrical installations.

4.1.6 A multifunction safety compliance analyser can determine the exact location of cable failure.

4.1.7 A multimeter can be used to detect installation damage that may affect cable performance.

4.1.8 Switchyards are generally classified according to groupings.

4.1.9 If a test fails, all wiring should be repeated.

4.1.10 An earth loop impedance tester is connected at the main switch.

(10 × 1) (10)

4.2 Describe the following plans:

4.2.1 Cross-sectional plans

4.2.2 Floor plans

(2 × 2) (4)

4.3 Between two plans listed in QUESTION 4.2 above, which one is more helpful for electrical installations and why? (2)

4.4 Before any test is performed using measuring instruments, test leads must be checked.

Name TWO items that the test leads must be checked for. (2)

4.5 Briefly explain why jewellery and metal objects must be removed when working with electricity. (2)

[20]

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QUESTION 5

5.1 Explain the procedure when using fault circuit indicators during fault finding on a cable. (2)

5.2 A cable supplying a lathe machine is suspected to be damaged.

Name TWO factors that could cause damage to a cable. (2)

5.3 List THREE reasons for transformer overheating. (3)

5.4 The contacts of a contactor for an electric motor are chattering.

State TWO possible causes and a remedy for each. (4)

5.5 State FOUR activities during normal maintenance on an electric machine. (4)
[15]

QUESTION 6

6.1 Briefly describe a *solar array*. (2)

6.2 Ms Tlali wants to connect some of the appliances at her house to a solar system to save some money.

The table below provides information on the appliances, their power usage and the hours they are in use. The cost per kW.h is R2,05.

Appliances	Power use (Watts)	Hours in use (per day)
8 Bulbs	8 each	5
Microwave	1 300	1
Geyser	1 500	3

Calculate the following:

6.2.1 Power consumption for each appliance in kW.h/day (6)

6.2.2 Total power consumption in kW.h/day (2)

6.2.3 The amount that Ms Tlali will be able to save in rand (2)

6.3 Explain the term *battery discharge depth* used for solar energy. (2)

6.4 Give ONE reason for connecting batteries in a battery bank either in parallel or in series. (1)
[15]

TOTAL: 100

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